

# A valuation obstruction for $3 \times 3$ magic squares of squares, with a machine-checked proof

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## Abstract

We study the open problem of whether a  $3 \times 3$  magic square whose nine distinct entries are perfect squares exists. We reduce the problem to a purely additive condition on the set  $D(e) = \{2xy : x^2 + y^2 = e^2\}$  attached to the center root  $e$ , and prove unconditionally that the center root of any such square must be divisible by at least two distinct primes  $p \equiv 1 \pmod{4}$ : for centers  $e = sp^a$  with a single such prime, the elements of  $D(e)$  have pairwise distinct  $p$ -adic valuations, which is incompatible with the required additive structure. All results, including the reduction, are formally verified in Lean 4 with mathlib. We also report an exhaustive computation showing no magic square of squares has center entry below  $10^{20}$ , and that Bremner’s fully magic square with seven square entries is, up to scaling, the only one with center root below  $10^9$ . Finally we describe a rigidity phenomenon that pins down the remaining two-prime case to explicit divisibility coincidences.

## 1 Introduction

Whether nine distinct perfect squares can form a fully magic  $3 \times 3$  square is a well-known open problem, popularized by Martin Gardner’s 1996 prize and studied extensively since; see Boyer’s survey and problem collection [1]. The strongest results to date are computational (searches over magic sums and over arithmetic-progression parameters) together with structural constraints on hypothetical solutions due to Morgenstern, Roberts, Underwood and others [1].

Throughout, a *magic square of squares* is a  $3 \times 3$  array of nine distinct perfect squares whose three rows, three columns and both diagonals share a common sum  $S$ . Summing the four lines through the center shows  $S = 3E$  where  $E = e^2$  is the center entry; we call  $e$  the *center root*.

## Results

**Theorem 1** (Reduction). *Let  $D(e) = \{2xy : x^2 + y^2 = e^2, 0 < x < y\}$ . A magic square of squares with center  $e^2$  exists if and only if there are  $u, v$  with  $u, v, u + v, u - v \in D(e)$  (all nonzero,  $u \neq v$ ); the square’s entries are then  $e^2, e^2 \pm u, e^2 \pm v, e^2 \pm (u + v), e^2 \pm (u - v)$ .*

**Theorem 2** (Main theorem). *If the center root  $e$  is divisible by at most one prime  $p \equiv 1 \pmod{4}$  (to any power, and with arbitrary cofactor free of such primes), then no magic square*

of squares with center  $e^2$  exists. Equivalently: the center root of any magic square of squares is divisible by at least two distinct primes  $\equiv 1 \pmod{4}$ .

**Theorem 3** (Computations). *There is no magic square of squares with center entry  $e^2 < 10^{20}$ . Moreover, every fully magic  $3 \times 3$  square with exactly seven square entries and center root  $e < 10^9$  is an integer scaling of Bremner's example with  $e = 425$ .*

All statements of Theorems 1 and 2, including Fermat's theorem that three squares in arithmetic progression have common difference divisible by 24, are formally verified in Lean 4 over mathlib; the library (thirteen files, no unproven assumptions) accompanies this paper.

## 2 The reduction

Every  $3 \times 3$  magic square has the form  $c \pm \{0, u, v, u + v, u - v\}$  arranged as

$$\begin{pmatrix} c+u & c-u-v & c+v \\ c-u+v & c & c+u-v \\ c-v & c+u+v & c-u \end{pmatrix},$$

in which all eight line sums equal  $3c$  identically. Thus magic is free and the entire problem is the squareness of the nine cells.

**Lemma 4.** *For  $c = e^2$ , both  $c - d$  and  $c + d$  are squares if and only if  $d \in D(e) \cup \{0\}$  (up to sign). Indeed if  $x^2 + y^2 = e^2$  then  $c - 2xy = (x - y)^2$  and  $c + 2xy = (x + y)^2$ ; conversely if  $c - d = A^2$  and  $c + d = B^2$  then  $A^2 + B^2 = 2e^2$ ,  $A \equiv B \pmod{2}$ , and  $x = \frac{B-A}{2}$ ,  $y = \frac{B+A}{2}$  satisfy  $x^2 + y^2 = e^2$ ,  $2xy = d$ .*

Theorem 1 follows by applying Lemma 4 to the four center lines. (Distinctness of the nine entries corresponds to  $u, v, u \pm v \neq 0$  and  $u \neq v$ .)

## 3 Proof of the main theorem

Fix a prime  $p \equiv 1 \pmod{4}$ , write  $p = \pi \bar{\pi}$  in  $\mathbb{Z}[i]$  with  $\pi = A + Bi$ ,  $A^2 + B^2 = p$ , and let  $e = sp^a$  with  $a \geq 0$  and  $s$  free of primes  $\equiv 1 \pmod{4}$ .

**Lemma 5** (Rigid part). *Every  $z \in \mathbb{Z}[i]$  with  $N(z) = s^2$  is a unit multiple of  $s$ .*

*Proof.* Induct on the least prime factor  $q$  of  $s$ . If  $q \equiv 3 \pmod{4}$  then  $q$  is inert;  $q \mid N(z) = z\bar{z}$  gives  $q \mid z$ , and we divide. If  $q = 2$ , then  $1 + i$  divides  $z$  twice (as  $2 \mid N(z)$  twice), and  $(1 + i)^2 = 2i$  is 2 times a unit.  $\square$

**Lemma 6** (Classification). *Every  $z \in \mathbb{Z}[i]$  with  $N(z) = s^2 p^n$  equals  $us^j \pi^j \bar{\pi}^{n-j}$  for a unit  $u$  and some  $0 \leq j \leq n$ .*

*Proof.* Induct on  $n$ : for  $n \geq 1$ ,  $\pi \mid p \mid z\bar{z}$ , so  $\pi \mid z$  or  $\bar{\pi} \mid z$ ; divide and recurse, finishing with Lemma 5.  $\square$

**Lemma 7** (Bridge). *For every  $m \geq 1$ ,  $p \nmid \text{Im}(\pi^m)$ .*

*Proof.* Pick  $\omega \in \mathbb{Z}/p$  with  $\omega^2 = -1$  (possible as  $p \equiv 1 \pmod{4}$ ) and consider the ring maps  $\varphi_{\pm} : \mathbb{Z}[i] \rightarrow \mathbb{Z}/p$ ,  $i \mapsto \pm\omega$ . If  $p \mid \text{Im}(\pi^m)$  then, since  $\text{Re}(\pi^m)^2 + \text{Im}(\pi^m)^2 = p^m$ , also  $p \mid \text{Re}(\pi^m)$ , so  $\varphi_{\pm}(\pi)^m = 0$ ; as  $\mathbb{Z}/p$  is a field,  $\varphi_{\pm}(\pi) = 0$ , i.e.  $A \pm B\omega \equiv 0$ . Adding and subtracting gives  $p \mid A$ ,  $p \mid B$ , contradicting  $A^2 + B^2 = p$ .  $\square$

**Lemma 8** (Representation structure). *Let  $x^2 + y^2 = e^2 = s^2 p^{2a}$  with  $xy \neq 0$ . Then there are  $1 \leq t \leq a$  and  $\varepsilon \in \{\pm 1\}$  with*

$$2xy = \varepsilon p^{2(a-t)} (s^2 \text{Im} \pi^{4t}), \quad p \nmid s^2 \text{Im} \pi^{4t}.$$

*In particular  $\nu_p(2xy) = 2(a-t)$ , and distinct elements of  $D(e)$  have pairwise distinct  $p$ -adic valuations  $\{0, 2, \dots, 2a-2\}$ .*

*Proof.* Write  $z = x + yi$ ; by Lemma 6,  $z = u s \pi^j \bar{\pi}^{2a-j}$ . Then  $2xy = \text{Im}(z^2)$  and  $z^2 = u^2 s^2 \pi^{2j} \bar{\pi}^{2(2a-j)}$  with  $u^2 = \pm 1$ . If  $j = a$  the product is real and  $xy = 0$ . Otherwise, with  $t = |j-a|$ ,  $\pi^{2j} \bar{\pi}^{2(2a-j)} = p^{2(a-t)} \cdot \pi^{\pm 4t}$ , and  $\text{Im}(\pi^{4t}) = -\text{Im}(\bar{\pi}^{4t})$ ; absorb signs into  $\varepsilon$ . Lemma 7 and  $p \nmid s$  give the non-divisibility.  $\square$

*Proof of Theorem 2.* Suppose  $u, v, u+v, u-v \in D(e)$  as in Theorem 1, and give each the form of Lemma 8 with parameters  $(t_i, \varepsilon_i)$  and constants  $c_t = s^2 \text{Im} \pi^{4t}$ . If  $t_1 = t_2$  then  $u = \pm v$ , excluded. Otherwise  $\nu_p(u) \neq \nu_p(v)$ ; writing  $u \pm v = p^{\min} \cdot (\text{bracket})$  where the bracket is  $\varepsilon_i c_{t_i} + (\text{multiple of } p)$ , the bracket is prime to  $p$ . Comparing with the exact forms of  $u+v$  and  $u-v$  and using uniqueness of exact  $p$ -power extraction,  $2(a-t_3) = 2(a-t_4) = \min$ , so  $t_3 = t_4$  and  $u+v = \pm(u-v)$ , forcing  $u = 0$  or  $v = 0$ . Contradiction.  $\square$

*Remark 9.* The counting argument (four center lines require four representations) already forces two distinct primes or  $p^4 \mid e$ ; the valuation argument closes the prime-power loophole entirely and is what we formalize.

## 4 Computations

Enumerating candidate centers directly through Theorem 1 is far cheaper than enumerating magic sums: factoring  $e$  in windows and generating  $D(e)$  from the Gaussian factorization checks all  $e < 10^{10}$  (center entries to  $10^{20}$ ) in minutes on a desktop; no configuration  $u, v, u \pm v \in D(e)$  exists in that range. The same machinery classifies near-misses: configurations with two of the four differences in  $D(e)$  and the other two “half in” (one square cell each) are exactly the fully magic squares with seven square entries, of which Bremner’s  $e = 425$  example was the only one known. Scanning all  $e < 10^9$  finds 2,352,941 such configurations — every single one an integer scaling of  $e = 425$ , and none with eight square entries.

## 5 The minimal two-prime case

**Theorem 10.** *There is no magic square of squares whose center root is  $spq$ , where  $p \neq q$  are primes  $\equiv 1 \pmod{4}$ , each to the first power, and  $s$  is free of such primes.*

*Proof sketch.* Here  $D(e)$  has exactly four elements: the interior conjugate pair  $m, m' = s^2 |\text{Im}(\pi^4 \chi^{\pm 4})|$  and the axis elements  $X = s^2 q^2 |\text{Im} \pi^4|$ ,  $Y = s^2 p^2 |\text{Im} \chi^4|$ . Since  $u, v, u+v, u-v$

are four distinct elements of  $D(e)$ , they exhaust it. Write  $\pi^4 = R_A + I_A i$ ,  $\chi^4 = R_B + I_B i$ ; recall  $R_A, R_B$  are odd,  $4 \mid I_A, I_B$ , and by Lemma 7  $p \nmid I_A$ ,  $q \nmid I_B$  (and  $I_A, I_B \neq 0$ ). Enumerate the placements of  $\{X, Y\}$  among  $\{u, v, u+v, u-v\}$ :

(1)  $\{u, v\} = \{X, Y\}$ . Then  $\{u \pm v\} = \{m, m'\}$  and the conjugate-pair identities force  $\{u, v\} = \{s^2 |R_B I_A|, s^2 |I_B R_A|\}$ . Matching against  $X, Y$  and multiplying the two resulting equations gives  $|R_A R_B| = p^2 q^2$ ; but  $|R_A| < p^2$  and  $|R_B| < q^2$  strictly (as  $I_A, I_B \neq 0$ ), a contradiction. The crossed matching forces  $|R_A| = p^2$  outright, equally impossible.

(2)  $\{u, v\} = \{m, m'\}$ . Then  $\{u \pm v\} = \{X, Y\}$  and  $u+v, u-v \in \{2s^2 |R_B I_A|, 2s^2 |I_B R_A|\}$ ; matching yields either  $2|R_B| = q^2$  or  $4|R_A R_B| = p^2 q^2$ , both impossible by parity.

(3) *mixed placements*. One axis element lies in  $\{u, v\}$ , the other in  $\{u \pm v\}$ . Each of the 64 sign/placement sub-cases yields two linear equations in  $W = R_A I_B$ ,  $Z = I_A R_B$ ; solving and substituting into the norm relations  $(W/I_B)^2 + I_A^2 = p^4$ ,  $(Z/I_A)^2 + I_B^2 = q^4$  leaves a difference that factors, in every sub-case, into terms from the list

$$(I_A q^2 \pm I_B p^2), \quad (2I_A q^2 \pm I_B p^2), \quad (I_A q^2 \pm 2I_B p^2), \quad 3I_A q^2, \quad 3I_B p^2.$$

The all-positive combinations cannot vanish; the difference combinations vanish only if  $I_A q^2 \in \{I_B p^2, 2I_B p^2\}$  or  $I_B p^2 = 2I_A q^2$ , each of which equates a quantity prime to  $p$  with one divisible by  $p^2$  (Lemma 7). Hence no sub-case admits a solution.  $\square$

*Remark 11.* Bremner's seven-square-entry example has center root  $425 = 5^2 \cdot 17$ , just outside the hypothesis (exponent 2): consistent with Theorem 10, the near-miss lives at the smallest center type the theorem does not reach.

## 6 Center roots $sp^2q$ : Bremner's type

**Theorem 12.** *There is no magic square of squares whose center root is  $sp^2q$ , where  $p \neq q$  are primes  $\equiv 1 \pmod{4}$  and  $s$  is free of such primes.*

Here  $|D(e)| = 7$ : writing  $\pi^4 = R_A + I_A i$ ,  $\chi^4 = R_B + I_B i$  and  $P = p^2$ ,  $Q = q^2$ , the elements are (up to the global factor  $s^2$ )

$$P |\operatorname{Im}(\pi^4 \chi^{\pm 4})|, \quad |\operatorname{Im}(\pi^8 \chi^{\pm 4})|, \quad PQ |I_A|, \quad Q |\operatorname{Im} \pi^8|, \quad P^2 |I_B|.$$

All seven are *linear* in  $(R_B, I_B)$  over  $\mathbb{Q}(R_A, I_A, P, Q)$ . The four differences  $u, v, u+v, u-v$  occupy four distinct elements; for each of the 840 placements and each resolution of the absolute values (6,720 linear systems in all), solving the two placement equations for  $(R_B, I_B)$  and substituting into  $R_B^2 + I_B^2 = Q^2$  leaves one polynomial relation in  $(R_A, I_A, P, Q)$ , which must vanish. A computer-assisted but fully rigorous sweep (scripts accompany this paper) shows every relation is impossible:

- 339 of the 343 distinct relation classes die by combinations of: parity ( $R_A, R_B, P, Q$  odd,  $8 \mid I_A, I_B$ ); the  $p$ -adic clash ( $\nu_p$  of the two sides differ, using  $p \nmid R_A I_A$  and Lemma 7); congruence obstructions modulo 16, 32 and small odd moduli on the curve  $R_A^2 + I_A^2 = P^2$ ; the strict size pinch  $|R_A|, |I_A| < P$ ; and elimination of  $I_A$  via the norm followed by factorization. The finitely many exceptional primes ( $\ell \in \{5, 13, 17\}$ , arising from small constants in the eliminants) are closed by exact evaluation at the actual  $\pi^4$ .

- 4 classes reduce to homogeneous quintics/sextics in  $(R_A, P)$ ; a rational root  $R_A/P$  in lowest terms would need  $P = p^2$  to divide a leading coefficient in  $\{\pm 8, \pm 16, \pm 32, \pm 40\}$  — impossible.
- 4 classes are tautological (the norm residual vanishes identically); there the solved values are  $(R_B, I_B) = \pm(Q/P)(R_A, -I_A)$ , and integrality forces  $p^2 \mid q^2$ .

The 256 distinct determinants of the linear systems are proven nonzero by the same mechanisms, so no singular case escapes.  $\square$

*Remark 13.* Bremner’s fully magic square with seven square entries has center root  $425 = 5^2 \cdot 17$ , of exactly this type: Theorem 12 shows the missing two entries can never be repaired at *any* center of that shape. Since the seven  $D$ -elements are linear in  $(R_B, I_B)$  for every  $b = 1$  exponent pattern  $sp^aq$ , the same pipeline applies to all  $(a, 1)$  cases; the case count grows with  $a$ , and a uniform-in- $a$  argument is the natural next objective.

**Theorem 14.** *There is no magic square of squares whose center root is  $sp^3q$  ( $p \neq q$  primes  $\equiv 1 \pmod{4}$ ,  $s$  free of such primes).*

For  $a = 3$  the difference set has ten elements, still linear in  $(R_B, I_B)$ ; the sweep comprises 64,512 leaf systems in 3,767 relation classes. All 1,271 nonsingular classes die by the Theorem 12 mechanisms; the 2,496 singular classes (rank-deficient placements, which first appear at  $a = 3$  through the three constant axis elements) die through their consistency relations by the same kills; and all 1,080 determinant classes are proven nonzero, so no singular locus escapes. The certificates for  $(a, b) = (1, 1), (2, 1), (3, 1)$  share one pipeline, and every two-prime exponent pattern  $(a, b)$  is decidable in this sense — for  $b \geq 2$  the leaf systems become zero-dimensional quadratic rather than linear, since  $\chi^8, \chi^{12}, \dots$  are polynomials in  $(R_B, I_B)$ . The open problems are uniformity in the exponents and the  $\geq 3$ -prime landscape, where the leaf systems acquire positive dimension.

## 7 The two-prime frontier

### Toward uniformity in the exponent

The certificates for  $(a, b) = (1, 1), (2, 1), (3, 1)$  suggest a uniform theorem for all  $sp^aq$ . Abstracting the levels of  $D(e)$  into free circle variables linked by a chain map (the enriched relaxation), 95.8% of the 9,216 uniform two-level leaf systems die by mechanisms independent of  $a$  — positivity pinches, parities,  $p$ -adic patterns and congruence obstructions. The residue is exactly 24 relations, independent of  $a$ , involving two  $\pi$ -powers  $\pi^{4j}, \pi^{4(j+\Delta)}$ . A complete weight-regime analysis shows that their phase-aligned leading forms cancel identically: the obstruction survives every first-order  $p$ -adic tool. The residue was then resolved further by the *exact orbit substitution*: under the two  $p$ -adic embeddings,  $R_j - \omega I_j = \pi_-^{4j}$  exactly, so each relation becomes a finite Laurent polynomial with definite monomial valuations. Single-level templates die completely (unit leading coefficients; the tautological placements force  $\text{Im } \chi^4 = 0$ ; the  $p = 5$  shapes die by exact 5-adic evaluation over a full period). Two-level templates die except on six explicit loci  $\pi_+^{4\Delta} \equiv c \pmod{p}$ ,  $c \in \{\pm 1, \pm 2, \pm \frac{1}{4}\}$ ; since  $\pi_+^{4\Delta} = c$  exactly is impossible (norms), each locus kills all but finitely many  $j$  per  $(p, \Delta)$ , and effective bounds on  $v_p(\pi_+^{4\Delta} - c)$  —  $p$ -adic linear forms in logarithms — yield  $j \ll \log^2 \Delta$ : a complete

effective-finiteness program for the uniform theorem over all  $sp^aq$ , with only the Baker-bound bookkeeping, the resulting finite verification, and the three- and four-level chain templates outstanding.

### Three and four levels: the $Q$ -collapse and the leading pair

The three- and four-level chain templates are now closed to the same standard. A merged sweep (36,288 three-level and 31,104 four-level leaves) kills 95.7%, resp. 91.4%, by the zero-solution, imbalance, residual-unit and singular mechanisms. The residue collapses to 8, resp. 12, *deep shapes* — same-orientation interior elements plus one axis element — and for each of the twenty shapes a symbolic computation gives the decisive structural fact: writing the solved value as  $\chi^4 = QN/D$ , the numerator has  $Q$ -degree exactly one and the denominator is  $Q$ -free. Setting  $N = QA$ , the modulus condition  $|\chi^4| = Q^2$  eliminates  $q$  *entirely*: a deep-shape leaf survives if and only if

$$c = |A|^2 - |D|^2 = 0,$$

a single integer identity in the  $\pi$ -orbit algebra. Expanding  $c$  over its monomial valuation groups  $\gamma \pi^{P \cdot j} \bar{\pi}^{B \cdot j}$  (the substitution is exact, with no unit ambiguity), realness pairs the groups conjugately, and for *every* shape and *every* strong-separation ordering of the orbit indices the minimal-valuation set is exactly one conjugate pair with real coefficient  $\gamma \in \{\pm 1, \pm 2\}$  — a unit for every odd  $p$ . The pair sums to  $2\gamma p^{a_0} \operatorname{Re}(\bar{\pi}^{4M})$  and  $v_p(\operatorname{Re}(\bar{\pi}^{4M})) = 0$ , so  $v_p(c)$  is finite: *every deep-shape leaf whose orbit indices avoid an explicit finite union of valuation-tie hyperplanes is impossible, uniformly in  $p \geq 5$ .*

On the tie set, layering  $c$  by  $\pi$ -exponent reduces everything to integer *phase sums*  $S_e(j) = \sum_i \gamma_i (2A)^{B_i \cdot j}$  (here  $\pi = A + Bi$ ): survival requires  $p \mid S_\pi(j)$  and  $p \mid S_{\bar{\pi}}(j)$  simultaneously — a double multiplicative-order condition on  $2A \bmod p$ . A census over  $p < 200$ , orbit indices  $\leq 6$ , finds 1,806 such double-vanishing points among 362,880 (0.5%); exact evaluation shows  $c \neq 0$  at every one, with even escape depths  $v_p(c) - e_{\min} \in \{2, \dots, 20\}$  governed by the multiplicative order of  $2A$  — the special-value loci in exact arithmetic form. Independently, a direct sweep of the criterion over all twenty shapes ( $p < 200$ , indices  $\leq 4$ ,  $q < 500$ ; 3,236,352 exact checks) finds no survivor. The uniform program for  $sp^aq$  thus stands closed at one and two levels unconditionally, and at three and four levels outside the thin double-vanishing set, where the small-order degeneration awaits its own (expectedly self-similar) leading-pair analysis.

### The tie-structure dichotomy

The hyperplane residue itself has now collapsed. Certifying the valuation-group geometry by exact rational linear programming (a phase-one simplex over  $\mathbb{Q}$ , no floating point), one finds for *every* one of the twenty deep shapes: every  $\pi$ -weight form carries a single group; no three forms can simultaneously achieve the minimum; and exactly *one* pair of forms is jointly-minimal-feasible — each shape owns a single tie hyperplane. On it, conjugation symmetry makes the minimal  $\bar{\pi}$ -layer sum to  $(\gamma_1 + \gamma_2)t^e$  with  $t = 2A \bmod p$ . The certified coefficient pairs split the shapes in two:

- **Eight shapes have**  $(\gamma_1, \gamma_2) = (1, 2)$ : the layer is  $3t^e \not\equiv 0 \pmod{p}$  for every  $p \geq 5$ , so  $v_\pi(c)$  is finite always and  $c \neq 0$  — *these eight shapes are impossible unconditionally: all  $p \geq 5$ , all orbit indices, all  $q$ .*
- **Twelve shapes have**  $(\gamma_1, \gamma_2) = (-1, 1)$ : the layer vanishes identically, and the surviving locus is exactly the tie hyperplane, where the leading contribution is  $\pi^e \bar{\pi}^b (\bar{\pi}^{\Delta b} - 1)$  and lifting-the-exponent gives the exact escape depth  $v_\pi(\bar{\pi}^d - 1) + v_p(\Delta b/d)$  when  $d = \text{ord}_p(2A)$  divides  $\Delta b$  (and death otherwise).

A census over  $p < 200$ , indices  $\leq 6$  confirms the classification exactly: all 1,806 double-vanishing points lie in the twelve  $(-1, 1)$ -shapes, every one is the trivial cancellation (no mod- $p$  accidents anywhere), and  $c \neq 0$  at each.

In fact the  $(-1, 1)$  cancellation is *exact*: the tie  $P \cdot j = B \cdot j$  forces both weight coordinates of the pair to coincide, so the two terms annihilate as Gaussian integers and  $c$  equals the sum of the remaining groups on the locus. Running the leading analysis *recursively* — delete each exactly-cancelling pair, add its equality, and re-analyze on the sub-locus, terminating by dimension exhaustion (observed depth  $\leq 2$ ) — closes every branch of every shape except finitely many explicit *order loci* (about nineteen in total across the twelve shapes): ties of opposite unit coefficients in a single coordinate with distinct exponents  $b_1 \neq b_2$ . There the layer sum is  $\gamma(t^{b_1} - t^{b_2})$  with  $t = 2A \bmod p$ , zero precisely when  $\text{ord}_p(2A)$  divides  $b_1 - b_2$ , in which case the valuation deepens by exactly  $v_\pi(\bar{\pi}^{b_1 - b_2} - 1)$  — a Fermat-quotient (Wieferich-type) quantity. Survival would require this deepening to outrun every remaining layer; the exact censuses refute it at every point in range. This is the complete structural characterization of the uniform  $sp^aq$  frontier at three and four levels: finitely many Wieferich-type loci per shape, and everything else proven impossible.

## Both exponents two: the $(2, 2)$ program

For  $e = sp^2q^2$  the set  $D(e)$  has exactly twelve elements with a clean two-dimensional grading,

$$D(e) = \{ p^{4-2\alpha} q^{4-2\beta} s^2 \text{Im}(\pi^{4\alpha} \chi^{\pm 4\beta}) : (\alpha, \beta) \in \{0, 1, 2\}^2 \setminus \{(0, 0)\} \}.$$

Of the 95,040 placement-and-sign leaves, 93.5% die by the *two-prime grading lemma*: a relation whose minimal  $p$ -layer (or  $q$ -layer) holds a single term is impossible, since that term is a  $p$ -unit (modulo  $\pi$ ,  $\text{Im}(\pi^{4\alpha} \chi^{4\beta})$  reduces to a unit multiple of  $\bar{\pi}^{4\alpha} \bar{\chi}^{4\beta}$ ) standing against terms divisible by two more powers of  $p$ . The 6,208 residual leaves collapse to 216 distinct three-term relations, and an exact sweep over every ordered pair of primes  $p \neq q \equiv 1 \pmod{4}$  below 1000 (39,234,560 checks) finds none satisfiable: *no magic square of squares has center root  $sp^2q^2$  with  $p, q < 1000$ , for any  $s$ .* Beyond the sweep, per-factor certificates close a majority of the residual structure *unconditionally*: each relation factors over  $\mathbb{Q}$ , and a factor is proven nonzero by parity (always-odd integers), by a *unit side* (its reduction modulo  $\pi$  or  $\chi$  factors into unit atoms times a nonzero constant  $c$ , so no prime beyond  $|c|$  ever divides it — and  $p \nmid F$  alone gives  $F \neq 0$ ), or by a *double pinch* (each side reduces to units times a single small core  $2\text{Re}(\chi^2)$  or  $2\text{Im}(\chi^2)$ , strictly bounded by twice the opposite prime and nonzero for an odd prime; the two prime orderings then close both cases with no wedge). Working in the full coordinates  $\chi = C + Di$  sharpens this again: the cyclotomic cores factor further ( $\text{Re } \chi^4 = (C^2 - 2CD - D^2)(C^2 + 2CD - D^2)$ ), a prime dividing a product divides a

single core, and cores of half-degree  $\frac{1}{2}$  (such as  $C \pm D \leq \sqrt{2q}$ ) bound one prime by roughly the square root of the other; chaining the two sides bounds *both* primes. All 110 relations of this type have explicit bounds at most 144 — inside the exhaustive sweep — so with the earlier certificates 134 of the 216 relations are dead and 98.35% of *all* 95,040 placement leaves are closed unconditionally. The 1,572 surviving leaves live in 82 relations: 18 reduce to explicit Pell-band equations (core =  $\pm kp$  inside a constant prime-ratio band), 64 carry genuine exponent- $\geq 2$  wedges, and 24 have a vanishing first-order side. For three primes the same analysis profiles all 600 residual relations: 116 (those with all three side exponents  $\leq 1$ ) are impossible outside constant prime-ratio bands, and the remainder confine any survivor to explicit power-law prime families ( $r \sim p^2$ -type rays of the associated log-inequalities) — a complete geometric characterization of the three-prime frontier.

All thirty-two pure double-pinch relations are now *machine-checked dead* in Lean 4 with no **sorry**. The mechanism is packaged as a reusable toolkit: `pi_core_disj` proves that  $\pi \mid (\varepsilon_2 q^2 + \varepsilon_3 \bar{\chi}^4)$  forces  $p \mid 2(C^2 - D^2)$  or  $p \mid 4CD$ ; `sq_core_disj` is the symmetric  $\chi$ -side statement; and `double_pinch_finish` shows any combination of the two core disjunctions is absurd by the strict two-ordering pinch (trichotomy on  $p$  versus  $q$ ). The thirty-two relations organize into four slot-triple families, each formalized by a parametric theorem over all sign choices  $\varepsilon_i \in \{\pm 1\}$ , all odd prime pairs, and all Gaussian representations: vanishing makes the Gaussian value self-conjugate, each prime's side subtracts its manifestly divisible terms from the conjugated relation (one conjugation class per family requires regrouping the six-term identity  $G - \bar{G} = 0$  before the extraction), unit factors are stripped by primality, and the exposed integer cores feed the pinch. This closes the double-pinch layer of a verified partial Theorem H'; the remaining twenty-four unconditionally-killed relations carry parity and unit-side certificates whose always-odd cubic cofactors await the same treatment.

The residual band relations were probed to second order: on several hundred integer points of the band loci the relation values carry  $v_p = 1$  at 96% of points but  $v_p = 2$  at about 4% — the mod- $p^2$  coefficient vanishes on a thinner sub-locus of its own, and each valuation order exhibits its own exceptional set. No bounded-valuation certificate is available by these methods: the last fraction of the two-prime  $b = 2$  landscape is guarded by the same arbitrary-depth (Fermat-quotient-flavored) thinning as the deep order loci of § *Toward uniformity*, and its closure requires depth-uniform arithmetic input beyond first- and second-order valuations.

The grading lemma is exponent-independent, and its reach *grows* with the exponents: for general  $e = sp^a q^b$  the set  $D(e)$  is always  $\{p^{2a-2\alpha} q^{2b-2\beta} s^2 \operatorname{Im}(\pi^{4\alpha} \chi^{\pm 4\beta})\}$  of size  $((2a+1)(2b+1)-1)/2$ , and the census of leaves killed by the lemma reads 93.5% at  $(2, 2)$ , 96.2% at  $(3, 2)$ , 97.7% at  $(3, 3)$ , and 98.9% at  $(4, 4)$ , with the residual relation count growing only polynomially (216, 468, 1008, 3040). Thus the *entire* two-prime landscape reduces, uniformly in the exponents, to explicit doubly-balanced relation families carrying coupled-residue survival conditions.

### Three primes

Nothing in the grading argument is special to two primes. For  $e = spqr$  (three distinct primes  $\equiv 1 \pmod{4}$ ) the set  $D(e)$  consists of thirteen elements  $p^{2-2\alpha} q^{2-2\beta} r^{2-2\gamma} s^2 \operatorname{Im}(\pi^{4\alpha} \chi^{\pm 4\beta} \psi^{\pm 4\gamma})$  carrying a three-dimensional grading, and the lone-minimal-layer kill applies in any of the three gradings (68.4% of the 137,280 leaves). An exact sweep of the 43,392 residual leaves over *every* ordered triple of distinct such primes below 120 (94,768,128 checks) finds none satisfiable: *no magic square of squares has center root  $spqr$  with  $p, q, r < 120$ , for any  $s$*

— the first exclusion in the  $\geq 3$ -prime landscape. The mixed shape  $e = sp^2qr$  behaves the same way ( $|D| = 22$ , 86.9% of 1,404,480 leaves grading-killed, 92,413,440 exact checks, no survivor below 75). The grading lemma itself is machine-checked: `im_pipow_mul_not_dvd` in the accompanying Lean library proves, with no `sorry`, that for  $p = a^2 + b^2$  prime,  $k \geq 1$ , and any Gaussian integer  $W$  with  $p \nmid N(W)$ ,  $p$  never divides  $\text{Im}((a + bi)^k W)$ . Four primes fall the same way: for  $e = spqrt$  ( $|D| = 40$ , four gradings), 81.2% of the 17,546,880 leaves die structurally and 396,656,640 exact checks exclude every quadruple from  $\{5, 13, 17, 29, 37\}$  — no center root  $spqrt$  with all primes below 40. The “positive-dimensional wall” blocked only the solved-form pipeline; the raw-relation grading passes through it, and the same balanced-family characterization governs any number of primes.

For  $e = sp^aq^b$  with two distinct primes  $\equiv 1 \pmod{4}$ , the valuation pairs  $(\nu_p, \nu_q)$  organize  $D(e)$  into *conjugate pairs*  $\{d_{j,k}, d_{j,-k}\}$  of multiplicity exactly two off the axes, and axis singletons. If  $\{u + v, u - v\}$  is a conjugate pair  $\{d, d'\}$  arising from  $A = \pi^{4j}$ ,  $B = \chi^{4k}$ , then

$$d + d' = 2 \text{scale} \cdot |\text{Re } B \text{ Im } A|, \quad d - d' = 2 \text{scale} \cdot |\text{Im } B \text{ Re } A|,$$

so  $u$  and  $v$  are *completely determined* — and their valuation pair coincides with that of  $\{d, d'\}$ , forcing  $u, v \in \{d, d'\}$ , which is absurd. Consequently every hypothetical two-prime solution requires an explicit divisibility coincidence ( $q^t \mid \text{Re } \pi^{4j}$ ,  $p^t \mid \text{Re } \chi^{4k}$ , or a mod- $p$  or mod- $q$  cancellation in  $u \pm v$ ). These coincidences recur along arithmetic progressions of exponents, so closing them requires exact-magnitude input beyond valuations; we leave this as the principal open direction, noting that together with Theorem 2 it would force every solution’s center root into three or more primes  $\equiv 1 \pmod{4}$ .

## References

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